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Kidney Physiology and Complications in Normal Pregnancy

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KIDNEY PHYSIOLOGY IN NORMAL PREGNANCY

Kidney physiology changes significantly in pregnancy. Marked volume expansion and vasodilation result in alterations in systemic and renal hemodynamics and kidney volume, with physiologic urinary tract dilation being common (Fig. 44.1A). Renal plasma flow (RPF) increases by up to 80% and glomerular filtration rate (GFR) by 50%.[1] Kidney tubular function responds to these changes with alterations in water and electrolyte handling, leading to clinical changes including lower serum osmolality, gestational hyponatremia, and increased urinary glucose and protein excretion (Table 44.1).

ANATOMY

In pregnancy, there is progressive dilation of the renal pelvis, calyces, and ureters. These changes are thought to be due to a reduction in smooth muscle tone and peristalsis mediated by progesterone, in conjunction with mechanical compression of the ureters by the gravid uterus. Dilation of the ureters is estimated to occur in up to 80% of pregnancies and is more prominent on the right side due to dextrorotation of the uterus by the sigmoid colon and kinking of the ureter as it crosses the right iliac artery.[2] Ureteric dilation of up to 20 mm on the right and 8 mm on the left is consistent with the altered physiology and anatomy of pregnancy[3] (see Fig. 44.1). Distinction between gestational dilation and pathologic hydronephrosis can be challenging. A dilated ureter distal to the pelvic brim, abnormal resistive indices in the kidney, and an absence of ureteric jets in the supine position are suggestive of obstructive pathology.[4] The dilated collecting system can hold 200 to 300 mL of urine, promoting urinary stasis and increasing the risk for clinical infection in women with asymptomatic bacteriuria (ASB).

HEMODYNAMIC CHANGES

Systemic

Plasma volume increases until 32 to 34 weeks' gestation by as much as 1.25 L (Fig. 44.2). Despite an increase in red blood cells (RBCs), there is a disproportional increase in plasma, causing a physiologic anemia of normal pregnancy.[5] Cardiac output increases by 40% to 50% secondary to increased heart rate, stroke volume, and venous return.[6] Despite this, systemic blood pressure (BP) decreases as a result of reduced systemic vascular resistance (SVR) thought to be mediated by vasodilatory factors, including progesterone, relaxin, and nitric oxide (NO).

Kidney

Systemic vasodilation, increased arterial compliance and decreased vascular resistance, and increased cardiac output result in increased kidney perfusion and glomerular filtration measurable at as early as 4 weeks' gestation, peaking at 40% to 50% above prepregnancy values by the second trimester.[7] These changes result in decreased serum creatinine concentrations in pregnancy (Table 44.2). Systematic review data show that serum creatinine in pregnancy falls to 86% or less of the nonpregnant upper reference limit. This means that where the nonpregnant reference interval is 0.51 to 1.02 mg/dL (45–90 μmol/L), a serum creatinine of more than 0.87 mg/dL (>77 μmol/L) should be considered outside the normal range for pregnancy and prompt investigation for either acute kidney injury (AKI) or chronic kidney disease (CKD).[8,8a]

Mechanisms of Increased Glomerular Filtration Rate

The exact mechanisms behind increased GFR in pregnancy are incompletely understood. GFR is calculated as:

$$\mathrm{GFR} = (\Delta \mathrm{P} - \pi_{\mathrm{GC}}) \times \mathrm{Kf}$$

where ΔP is the net hydraulic pressure in the glomerulus, π_{GC} is the oncotic pressure in the glomerulus, and Kf is the glomerular ultrafiltration coefficient. As a result of plasma volume expansion, π_{GC} is decreased in pregnancy, contributing to the rise in GFR. Changes in hydraulic permeability and the surface area for filtration may cause minor changes in Kf. Despite increased RPF, there is no overall change in glomerular pressure because dilation of the pre- and postglomerular resistance vessels are equal.

Measuring Glomerular Filtration Rate

True GFR measurement in pregnancy has historically been achieved using inulin or creatinine clearance methods. Estimation of GFR in pregnancy is challenging because the Modification of Diet in Renal Disease (MDRD) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formulas significantly underestimate true GFR in pregnancy. Similarly, cystatin C concentrations do not correlate with other measures of GFR in pregnancy.[9] The absence of large-scale

Fig. 44.1 Hydronephrosis in Pregnancy. Intravenous urogram (IVU) at 36 weeks' gestation. Note bilateral hydronephrosis, more marked on the right side. In contemporary clinical practice, ultrasound is the first-line imaging modality in pregnancy, with a possible role for magnetic resonance imaging or limited IVU.

TABLE 44.1 Changes in the Mean Value of Some Common Indices During Pregnancy

	Nonpregnant	Pregnant
Hemoglobin (g/L)	135	>105
Hematocrit (%)	40	34
Serum sodium (mmol/L)	140	135
Serum creatinine, mg/dL (mmol/L)	0.76 (67)	0.59–0.63 (52–56)[a]
Serum osmolality (mOsm/kg)	285	270
Blood urea nitrogen (mg/dL)	12.7	9.0
Serum urea (mmol/L)	4.5	3.2
Serum uric acid, mg/dL (μmol/L)	4.0 (240)	3.2–4.3 (190–260)
Serum albumin (g/L)	40	30
pH	7.40	7.44
Arterial P_{O_2}, mm Hg (kPa)	100 (13.3)	112 (15.0)
Arterial P_{CO_2}, mm Hg (kPa)	40 (5.3)	26 (3.5)
Serum bicarbonate (mmol/L)	25	20
Systolic BP (mm Hg)	115	113–121[b]
Diastolic BP (mm Hg)	70	69–78[b]

[a]Data from Ogueh O, Brookes C, Johnson MR. A longitudinal study of the maternal cardiovascular adaptation to spontaneous and assisted conception pregnancies. *Hypertens Pregnancy*. 2009;28(3):273–289.
[b]Data from Green LJ, Mackillop LH, Salvi D, et al. Gestation-specific vital sign reference ranges in pregnancy. *Obstet Gynecol*. 2020;135(3):653–664.
BP, Blood pressure.

normative data for eGFR in pregnancy by any formulas means that it cannot be used as a measure of kidney function in pregnancy, despite ubiquitous use outside of pregnancy.[10] Serum creatinine concentrations therefore remain the only current method for the clinical assessment of kidney function during pregnancy.

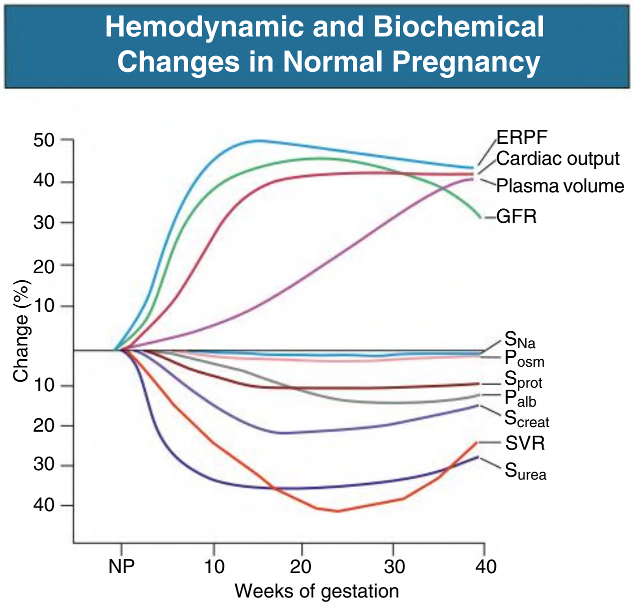

Fig. 44.2 Hemodynamic Alterations Induced by Normal Pregnancy. Increments and decrements in hemodynamic and biochemical parameters shown as percentage of change from nonpregnant baseline. *ERPF,* Effective renal plasma flow; *GFR,* glomerular filtration rate; *NP,* nonpregnant state; *Palb,* plasma albumin concentration; *Posm,* plasma osmolality; *S,* serum; *SVR,* systemic vascular resistance.

Renin-Angiotensin-Aldosterone System

In early gestation, vasodilation and decreased systemic vascular resistance activate the renin-angiotensin-aldosterone system to increase circulating concentrations of renin and angiotensin II. This mediates an aldosterone-dependent increase in plasma volume, thought to be essential for adequate uteroplacental blood flow. Maternal BP, however, does not increase, perhaps due to reduced responsiveness to the pressor effect of angiotensin II through altered sensitivity of arterial smooth muscle and downregulation of angiotensin receptors. Angiotensin,[1–7] which counterregulates angiotensin II and has vasodilatory and antihypertensive properties, is elevated in pregnancy and may contribute to vasodilation. Relaxin is a key mediator for enhanced nitric oxide signaling in pregnancy[11] and probably contributes to the vasodilated state of normal pregnancy. Relaxin is produced in the corpus luteum and the placenta in early gestation, leading to a peak in concentration at 6 weeks' gestation, returning to nonpregnant levels at 6 weeks postpartum. However, other mechanisms, possibly progesterone and nitric oxide, also have a role. This is evidenced by physiologic adaptation to early pregnancy after assisted conception, in which the corpus luteum is absent and relaxin is not detected.[12]

Sodium Handling and Osmoregulation

In early gestation, serum osmolality decreases by approximately 10 mOsm/kg to a new set point of about 270 mOsm/kg, with a reduction in serum sodium (see Table 44.1).[1] In pregnancy the physiologic vasopressin (AVP) release mechanisms of thirst are reset to recognize this new set point in osmolality, with β-human chorionic gonadotrophin (β-hCG) and relaxin hypothesized to play a role in this resetting. The gestational decrease in serum sodium concentration is thought to be secondary to systemic vasodilation, arterial underfilling, and subsequent release of AVP.[13] Although the natriuretic effects of increased GFR and elevated progesterone promote sodium excretion, this is balanced by increased aldosterone and angiotensin II directly increasing sodium reabsorption, increased deoxycorticosterone promoting

TABLE 44.2 Expected Normal Serum Creatinine Values According–Gestational Week: Derived From 244,866 in Pregnant Women With Uncomplicated Pregnancies

	SERUM CREATININE MG/DL (μMOL/L)		
Gestational Week	50th Percentile	75th Percentile	95th Percentile
4–10	0.54–0.67 (48–59)	0.6–0.72 (53–64)	0.69–0.85 (61–75)
10–32	0.50–0.54 (44–48)	0.55–0.6 (49–53)	0.66–0.69 (58–61)
32–42	0.51–0.62 (45–55)	0.57–0.71 (50–63)	0.68–0.92 (60–81)

TABLE 44.3 Antinatriuretic and Natriuretic Factors Influencing Sodium Excretion During Pregnancy

Antinatriuretic	Natriuretic
Aldosterone	Increased glomerular filtration rate
Angiotensin II	Progesterone
Estrogen	Atrial natriuretic peptide
Deoxycorticosterone	Nitric oxide
Epithelial sodium channel	Prostaglandins
Supine posture	
Upright posture	
Decreased blood pressure	
Increased urethral pressure	
Placental shunting	

sodium retention, and upregulation of the epithelial sodium channel (ENaC).[14] Although there is a gradual gain in total body sodium during pregnancy of about 900 to 1000 mmol, there is greater water gain (Table 44.3).

Potassium

Despite increased aldosterone and mild alkalosis in pregnancy, hypokalemia does not occur, and typical serum potassium concentrations in pregnancy are at the lower limit of the nonpregnant reference range. This may be partly due to the antimineralocorticoid effect of progesterone, as well as the capacity of progesterone to inhibit potassium excretion independent of antimineralocorticoid effects.

Uric Acid

Serum uric acid concentrations fall early in pregnancy before a return toward nonpregnant concentrations during late gestation (see Table 44.1). By 37 weeks' gestation, serum uric acid can reach nonpregnant levels. Gestational changes in the handling of uric acid by the kidney include increased clearance secondary to gestational hyperfiltration in conjunction with reduced fractional reabsorption. Other factors include dietary intake of purines and maternal and fetoplacental metabolism of purines; gastrointestinal excretion may also contribute.

Acid-Base

In pregnancy there is typically a chronic mild respiratory alkalosis secondary to an increase in tidal volume that lowers arterial carbon dioxide tension. Increased minute ventilation probably results from direct stimulation of the respiratory centers by progesterone and occurs without a change in respiratory rate.[15]

Urine Protein

There is a rise in urine protein excretion in normal pregnancy. The mechanisms and clinical implications of proteinuria in pregnancy are discussed later.

Glucose

Glycosuria in pregnancy occurs secondary to increased glucose filtration coupled with decreased tubular reabsorption. Once the filtered load of glucose has reached the maximum resorptive capacity of the proximal tubule, there is physiologic glycosuria, which can occur with normal plasma glucose concentrations.

Calcium

There is increased urinary excretion of calcium but also of magnesium, citrate, acidic glycoproteins, and nephrocalcin, which help prevent calcium stone formation associated with supersaturation.

KIDNEY DISEASE IN PREGNANCY

Urinary tract infection (UTI) is common in pregnancy. The most common causes of kidney injury in late pregnancy are preeclampsia and postpartum hemorrhage.

URINALYSIS AND MICROSCOPY

Many young women have urinalysis and urine microscopy performed for the first time during pregnancy, leading to the detection of hematuria, proteinuria, and pyuria either related to or coincidental to pregnancy.

Hematuria

Definition and Epidemiology

Nonvisible dipstick hematuria (microhematuria) is detected during pregnancy in about 20% of women[16] and in isolation, rarely signifies a disorder likely to influence pregnancy outcome. Clinically significant nonvisible hematuria is defined as three or more RBCs per high-power field on microscopic evaluation of two of three properly collected (clean catch, midstream) urine specimens,[17] or more than 2500 RBCs/mL. Nonvisible hematuria disappears in the majority of women after delivery, but if it is secondary to glomerular disease, it will persist, warranting further investigation.

Etiology and Outcome

Assessment of urinary RBC morphology differentiates dysmorphic red cells from isomorphic red cells. Outside of pregnancy, this may be used to help distinguish glomerular from nonglomerular sources of blood. However, dysmorphic red cells have been detected in women with severe preeclampsia in the absence of underlying glomerulonephritis.[18] In pregnancy, isomorphic hematuria is most likely to be caused by urinary tract infection. Isolated nonvisible hematuria (i.e., without proteinuria and with normal kidney function) has no adverse effect on pregnancy outcome with no measurable difference in gestational age at delivery, birth weight, gestational hypertension, or preeclampsia compared to pregnancies without nonvisible hematuria.[19]

Visible hematuria (macrohematuria) in pregnancy is most often the result of vaginal bleeding or hemorrhagic cystitis. Less common causes include renal calculi, renal arteriovenous malformations, renal vein thrombosis, polycystic kidneys, and rarely bladder or kidney neoplasms.

Differential Diagnosis

When nonvisible hematuria is found, a urine culture is required to exclude infection. If there is no proteinuria, and BP and serum creatinine values are normal, further investigations can be delayed until 3 months postpartum, when serologic tests and kidney ultrasound can be performed if hematuria persists. When there are significant numbers of dysmorphic urinary RBCs in pregnancy that persist postpartum, the most likely glomerular pathologies are thin basement membrane nephropathy, immunoglobulin A nephropathy, or lupus nephritis.

Treatment

There is no specific treatment for glomerular disease during pregnancy if kidney function and BP are normal, proteinuria is not clinically significant (see "Proteinuria" below), and lupus is excluded. Renin-angiotensin blockade is contraindicated in pregnancy due to fetal toxicity in the second and third trimesters. The treatment of UTI and calculi are discussed later.

Proteinuria

There is a physiologic increase in proteinuria in pregnancy due to increased permeability of the glomerular barrier in conjunction with impaired protein resorption in the proximal tubule. The established upper limit for normal protein excretion in pregnancy is based on data from small cohorts and expert opinion rather than an association with adverse pregnancy outcomes. The largest study of protein excretion measured by 24-hour urine collection included 270 women with mean protein excretion of 117 mg/day and an upper confidence interval limit of 259 mg/24 h.[20] This was extrapolated to produce the now established reference limit for pregnancy of 300 mg/24 h, compared with 150 mg/24 h outside of pregnancy. The increase is greater in twin pregnancy. The gestational increase in proteinuria can continue into the postpartum period and may take many months to resolve.[21]

Assessment

Proteinuria is most frequently detected in pregnancy by dipstick urinalysis, but this method is notoriously unreliable, with a significant proportion of false-positive and false-negative results.[22] Inaccuracy generally occurs due to the variable osmolality of a random urine specimen. Coexisting hematuria, alkaline urine (pH >7.0), and additives in the urine collection container also contribute to false positive tests. False-negative results may occur with acidic urine, low specific gravity (<1.001), and high salt concentration. In clinical practice, a dipstick result is sufficiently sensitive for the absence of significant proteinuria when testing is negative and for the detection of pathologic proteinuria when dipstick readings are 3+ (>3 g/L) or more. At intermediate levels, false-positive rates are as high as 71%. Detection is improved using an automated urinalysis device, thereby reducing observer error.[23]

Timed urine collections over 24 hours are limited by underestimation,[24] variability,[25] and the potential for treatment delay. Consequently, they have been superseded in clinical practice by spot quantification using a ratio of urinary protein or albumin to creatinine (uPCR or uACR, respectively).

Differential Diagnosis

Proteinuria quantified as uPCR greater than 30 mg/mmol (0.30 mg/mg, 300 mg/g) and uACR greater than 8 mg/mmol have high sensitivity and specificity for the detection of clinically significant proteinuria in pregnancy.[26] If detected before 20 weeks' gestation, preexisting kidney disease should be considered. Glomerular disease may present for the first time in pregnancy, or pregnancy may unmask kidney disease secondary to systemic disorders such as diabetes mellitus, systemic lupus, or chronic hypertension. The course of pregnancy in women with prepregnancy proteinuria is discussed in Chapter 45.

Gestational proteinuria is isolated, de novo proteinuria in pregnancy, without features of preeclampsia. Thus, the diagnosis can only be made in retrospect after completion of pregnancy. The significance of this is not known. In the absence of CKD, it is not thought to be associated with adverse effects for fetus or mother,[27] though surveillance is warranted for the development of preeclampsia.

Though proteinuria is no longer mandatory for the diagnosis of preeclampsia where other clinical features exist, when preeclampsia does cause proteinuria, it arises after 20 weeks' gestation and requires simultaneous de novo BP higher than 140/90 mm Hg.[28] Once above the diagnostic threshold for preeclampsia, the level of proteinuria has not been consistently associated with adverse pregnancy outcomes and repeat quantification is not recommended,[28,29] as it does not guide management.

We limit investigation of isolated, normotensive, nonnephrotic-range proteinuria during pregnancy to measurement of serum creatinine, electrolyte, and albumin concentrations; antinuclear antibody (ANA) titers; and an ultrasound to confirm normal renal tract morphology. A kidney biopsy is not indicated unless it will change management. In many cases appropriate investigations can be delayed until the postpartum period. However, kidney biopsy may be indicated in the first and early second trimester, when the bleeding risks of biopsy are lower[30] and where a histologic diagnosis will determine appropriate treatment during pregnancy, for example, the immunosuppressive management of nephrotic syndromes and lupus nephritis.

Treatment

Nephrotic syndrome due to primary kidney disease (and not preeclampsia) is associated with adverse maternal and fetal outcomes including preeclampsia, AKI, infection, preterm delivery, intrauterine growth restriction, and fetal loss.[31,32] Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers are contraindicated in the second and third trimesters of pregnancy due to toxic effects on the fetal kidney leading to anuria, oligohydramnios, fetal limb contractures, craniofacial deformities, and pulmonary hypoplasia. Diuretics are prescribed and sodium restriction is implemented for severe, intractable maternal edema including pulmonary complications, as in these settings the clinical benefit outweighs the theoretical concern of reduced plasma volume and uteroplacental perfusion. There is no established role for the use of intravenous (IV) albumin in managing nephrotic syndrome during pregnancy.

Nephrotic syndrome is a significant risk factor for venous thromboembolism. However, there is no consensus on the definition of nephrotic syndrome in pregnancy, which is complicated by gestational changes in protein excretion and serum albumin concentrations, and physiologic peripheral edema. There is expert consensus that nephrotic range proteinuria (uPCR >300 mg/mmol or uACR >250 mg/mmol) accompanied by low serum albumin warrants thromboprophylaxis with low-molecular-weight heparin (in the absence of contraindications or anticipated delivery within 12 hours) in pregnancy and the immediate postpartum period, with risk reassessed at 6 weeks postpartum. There is also consensus, but insufficient evidence, that subnephrotic levels of proteinuria confer an increased risk of thrombosis. Nonnephrotic range proteinuria (uPCR >100 mg/mmol or uACR >30 mg/mmol) is therefore considered a risk factor for thrombosis, and prophylactic use of low-molecular-weight heparin should be considered when there are additional risk factors for thromboembolism.[33]

URINARY TRACT INFECTION

Definitions

Bacteriuria is the detection of a clinically significant quantitative count of bacteria in the urine, defined as more than 10^5 organisms per milliliter. If this phenomenon exists without symptoms, it is defined as asymptomatic bacteriuria (ASB). Cystitis defines infection in the presence of lower urinary tract symptoms such as frequency, dysuria, and strangury. Although more than 10^5 organisms/mL defines ASB, as few as 10^2 organisms/mL may be sufficient to diagnose cystitis if accompanied by pyuria and characteristic symptoms. In *acute pyelonephritis* there are generally more than 10^5 organisms/mL in the urine, in association with parenchymal bacterial infiltration causing upper urinary tract symptoms. Pyelonephritis is usually diagnosed clinically by fever and loin pain and may progress to systemic sepsis.

Epidemiology

ASB is estimated to affect 2% to 9% of all pregnant women. The prevalence is higher in women from lower socioeconomic groups and increases with age, parity, and coexistent genital tract infection. ASB is also more common in women with urinary tract abnormalities including reflux nephropathy and neurogenic bladder, in diabetic patients, and in women with previous UTIs. Between 1% and 2% of pregnancies are complicated by acute bacterial cystitis. The overall incidence of acute pyelonephritis in pregnancy is approximately 1%. Pyelonephritis is more common in pregnant women with urologic abnormalities or diabetes and more often affects the right kidney, probably because physiologic dilation of the ureter is greater on the right. It is thought that about 70% of women who develop acute pyelonephritis have preceding covert bacteriuria, but this is difficult to prove. Pyelonephritis is estimated to occur in up to 30% of women with ASB. When ASB is treated, the incidence of pyelonephritis in pregnancy is estimated to be reduced by more than 80%.

Pathogenesis

Certain host characteristics may increase the risk for UTI or pyelonephritis (see Chapter 53). Women, such as those who do not express the antibody to the O antigen of *Escherichia coli*, may be chronically colonized and have ASB that predates pregnancy. Pregnancy is a state of relative urinary tract stasis; the calyces, pelvices, and ureters dilate, particularly on the right, and this contributes to the risk of ASB developing into ascending acute pyelonephritis. The most common mechanism of infection is through the urethra from perineal bacteria. Box 44.1 lists common organisms causing UTI in pregnancy. Some strains of *E. coli* are particularly virulent and are associated with both ASB and pyelonephritis. They possess fimbriae, which enable the bacteria to attach themselves to the uroepithelial cells with pili, allowing them to ascend the urinary tract from the perineum. Infection with multiresistant organisms is increasingly common.

BOX 44.1 Organisms Typically Causing Urinary Tract Infection in Pregnancy

- *Escherichia coli* (>70% of infections)
- *Klebsiella* spp.
- *Proteus* spp. (particularly in diabetic women or urinary tract obstruction)
- Enterococci
- Staphylococci, especially *Staphylococcus saprophyticus*
- *Pseudomonas*

Clinical Manifestations

Most maternity units operate a policy of screening all pregnant women at least once for ASB, either by dipstick urinalysis or by urine culture. Because isolated pyuria on dipstick testing is very common in normal pregnancy due to contamination from vaginal secretions, primary urine culture is recommended over dipstick testing.

Pyelonephritis most commonly presents between 20 and 28 weeks' gestation. Not all women will describe preceding lower urinary tract symptoms. Pyelonephritis can present in pregnancy as acute abdominal pain and may be associated with preterm labor, which is hypothesized to be triggered by proinflammatory cytokines secreted in response to bacterial endotoxins. The diagnosis of pyelonephritis is usually made on clinical grounds. Definitive diagnosis requires positive urine culture, but treatment should not be delayed while this is awaited. *E. coli* is the most common infecting organism (>85% of cultures). Bacteremia is a common and usually transient complication of pyelonephritis, although sepsis can develop with sequelae including AKI, disseminated intravascular coagulation, and respiratory distress. Pyonephrosis and perinephric abscess are rare complications but should be suspected in women who do not show a clinical response to treatment. In the preantibiotic era, maternal mortality was 3% to 4%; death from pyelonephritis is now rare in high-income countries.

Treatment

Asymptomatic Bacteriuria

A Cochrane systematic review has indicated that treatment of ASB during pregnancy may reduce the incidence of pyelonephritis, low birth weight (<2500 g), and preterm delivery,[34] though confidence in the effect is limited given the low certainty of the evidence. Most available studies are historical with low methodological quality and do not necessarily reflect contemporary advances in diagnosis and treatment and accessibility to medical services. A more recent study randomized 85 women with ASB (out of 4283 screened) to treatment with either nitrofurantoin 100 mg twice daily for 5 days or placebo. This trial showed no association between ASB and preterm birth. Although there was an increased rate of pyelonephritis with untreated ASB compared with antibiotic use, the absolute risk of pyelonephritis in untreated ASB was low (2.4%). Although such findings question a universal screen-treat policy for ASB in pregnancy, there is consensus in high-income countries that more data are required before this routine and long-standing strategy is reconsidered.

There is no clear evidence that any particular antibiotic or dosage regimen is superior for treatment of ASB in pregnancy. In choosing treatment, the clinician must consider urine culture and susceptibility results, previous antibiotic use, and safety data for pregnancy. In most women, amoxicillin or cephalexin is first-line therapy. Nitrofurantoin is not teratogenic, though it is used with caution close to term or anticipated delivery because of a rare risk of neonatal hemolysis. Trimethoprim can be used in pregnancy but should be avoided in established folate deficiency, low dietary folate intake, and in women taking other folate antagonists. Many clinicians avoid trimethoprim in the first trimester because of its antifolate effects. There is insufficient evidence to compare the effectiveness of a single dose compared with longer treatment duration, or a 3-day with a 7-day course of treatment, and a 7-day treatment regimen is therefore usually recommended. A urine culture should be performed after completion of antibiotic treatment to check for eradication.

Without treatment, ASB will persist in 80% of women, and even with treatment, 20% will have persistent bacteriuria. Those with persistent colonization are difficult to treat, with eradication achieved in only 40% after a second course of antibiotics. Where eradication is not achieved, prophylactic antibiotics may be considered throughout pregnancy to prevent pyelonephritis and its consequences; however, no studies specifically address this situation.

Cystitis

Cystitis is managed in the same way as ASB with a follow-up urine culture to confirm eradication of infection.

Pyelonephritis

It is usual practice to admit pregnant women with pyelonephritis to hospital, although successful outpatient management for milder cases has been reported.[35] Initial treatment with a broad-spectrum antibiotic is recommended with review according to clinical response and culture results. Prophylactic antibiotics are then recommended until delivery to prevent recurrence and the risk of preterm labor. Periodic urine culture can be considered to monitor for antibiotic resistance. Kidney ultrasound may be performed to ensure normal renal tract anatomy and is indicated in those not responding to antibiotic treatment. If pyelonephritis persists despite adequate antibiotic therapy and pathologic urinary tract dilation is confirmed, percutaneous nephrostomy under ultrasound guidance may be indicated to definitively manage an infected, obstructed renal tract.

KIDNEY STONES

Epidemiology

Despite normal pregnancy being an ideal environment for kidney stone formation, the incidence of kidney calculi remains similar in pregnant and nonpregnant women, in the range of 1 in 188 to 1 in 4600 in cohort studies, with lower incidences in unreferred pregnant populations.[36]

Pathogenesis

Most stones in pregnancy are calcium stones. However, in pregnancy, calcium phosphate stones are most common, in contrast to calcium oxalate in the nonpregnant population. Struvite stones are the next most common, usually when the urinary tract is infected with organisms such as *Proteus* spp. Small proportions of kidney stones are formed from uric acid or cystine. Although pregnancy is a physiologic state of relative urinary stasis and increased calcium and uric acid excretion, this is balanced by enhanced excretion of inhibitors of stone formation, such as magnesium, citrate, and the glycoprotein nephrocalcin.

Clinical Manifestations

Although symptomatic stones during pregnancy are rare, they are a relatively common cause of nonobstetric abdominal pain in pregnancy. Kidney and bladder stones usually present in the second or third trimester with acute flank pain radiating to the groin or lower abdomen, with hematuria. However, clinical features of kidney calculi may be more difficult to interpret in pregnancy because frequent episodes of diffuse, poorly localized abdominal discomfort and lower urinary tract symptoms can occur in normal pregnancy. Some women with calculi have concomitant UTI. Increased serum calcium concentrations in pregnancy warrant exclusion of hyperparathyroidism, which can lead to adverse maternal and fetal outcomes in untreated women.

Pregnant women with kidney calculi are reported to have an increased risk for superimposed pyelonephritis and obstetric complications including preterm labor, hypertensive disorders, gestational diabetes, and cesarean delivery.[37] However, the true risk of kidney stones in pregnancy is difficult to determine from existing studies.

The diagnosis of kidney calculi in pregnancy is difficult because of the radiation exposure to the fetus with computed tomography (CT) imaging, which is the investigation of choice outside of pregnancy. The fetal radiation exposures from abdominal CT (and most other diagnostic imaging procedures in common use) is below the threshold at which there is risk of fetal death, malformation, growth restriction, or impaired mental development.[38] However, as the radiation exposure threshold that leads to an increased relative (though low absolute) risk of childhood cancer is unknown, alternative imaging modalities that do not use ionizing radiation should be used when possible. Ultrasound is therefore often the first-line investigation in pregnancy, although detection rates are lower than with CT, varying from 29% to 69%.[36] Ultrasound will also detect gestational dilation of the urinary tract, which can complicate the diagnosis. An absence of ureteric jets in the contralateral decubitus position may be helpful in the assessment of obstruction. Transvaginal ultrasound can be added to examine for distal ureteral stones. If symptoms persist and further diagnosis is required, magnetic resonance (MR) urography or limited intravenous urography (IVU) have been suggested. MR imaging findings include a signal void, perinephric or periureteral edema, and an abrupt ending of the ureter at the level of obstruction. Limited IVU has been reported to have better diagnostic accuracy (93%) compared with ultrasound (60%).[40]

Treatment

Approximately 64% to 84% of stones will pass spontaneously during pregnancy, and so initial management of kidney calculi is conservative, including appropriate hydration, antiemetics, analgesia, and antibiotics if infection is suspected. The woman should lie on her side, with the symptomatic side up, enabling relief of pressure on the ureter from the gravid uterus. The dose of nifedipine used for expulsion of stones outside of pregnancy is comparable to that used in pregnancy for both hypertension and tocolysis and so can be safely given if clinically indicated. Quantitation of urine calcium, uric acid, or other mineral excretion is not performed in pregnancy, as interpretation in the context of gestational change is not clear and specific pharmacologic agents are not used in pregnancy due to limited data. Such investigations can be completed postpartum.

Surgical intervention is considered only when stones cause persistent obstruction, deteriorating kidney function, intractable pain or infection. This is a rare situation in pregnancy. Percutaneous nephrostomy can be used as a temporizing measure until after delivery, although the likelihood of tubal blockage should be considered, and flushing and/or replacement may be required. Ureteroscopic stone removal is increasingly described in pregnancy, with complete stone removal achieved in 86% of 116 procedures.[41] Lithotripsy is contraindicated during pregnancy because of the presumed adverse effect of the shock waves on the fetus, with fetal damage and death reported in animal studies. Inadvertent exposure to extracorporeal shock wave lithotripsy has been reported, however, leading to a recommendation for further research on possible utility in pregnancy.[36] Rare cases of percutaneous nephrolithotomy are reported in pregnancy; however, prolonged fluoroscopy time and prone positioning mean that it is usually delayed until the postpartum period.[36] Women should be assessed for idiopathic hypercalciuria or other causes of kidney calculi after delivery. Clinicians should be aware that gestational calyceal and ureteral dilation may persist for months postpartum.

HYPERTENSION IN PREGNANCY

Definitions

Median BP falls to a nadir of 113/69 mm Hg at 18 to 19 weeks' gestation and peaks at 121/78 mm Hg at term with an upper reference limit (97th centile) of 136/86 mm Hg in midpregnancy and 144/95 mm Hg

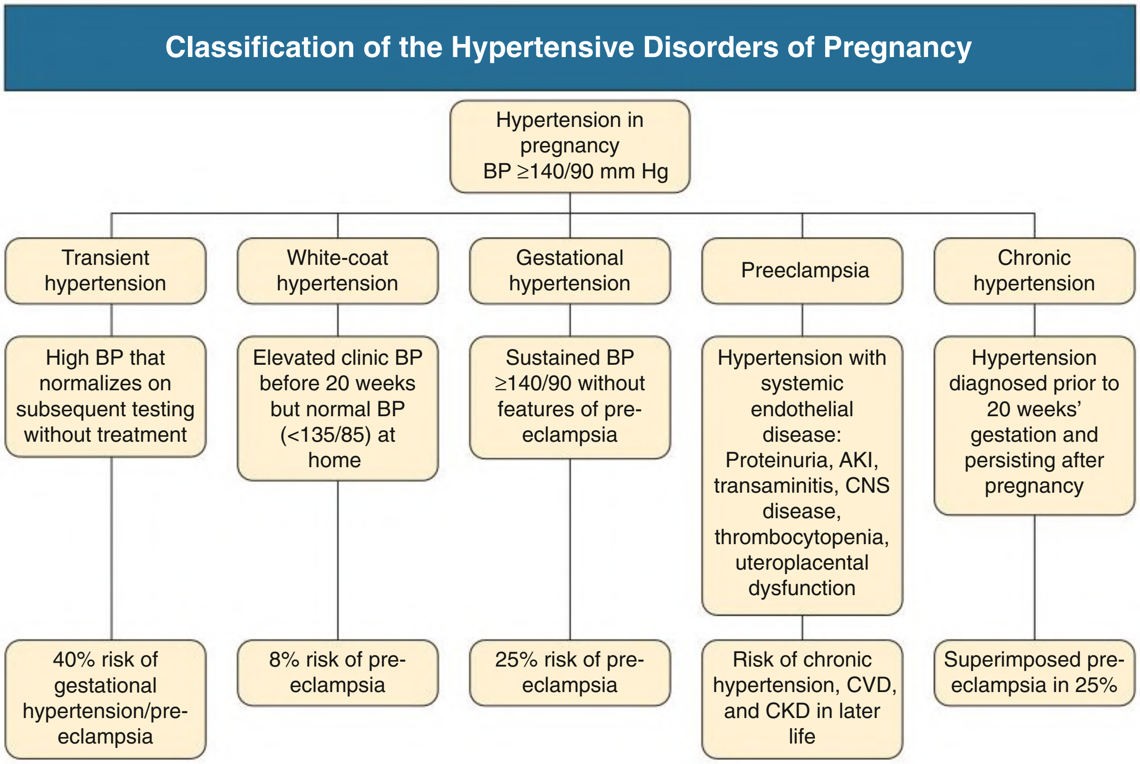

Fig. 44.3 Classification of the Hypertensive Disorders of Pregnancy. *AKI,* Acute kidney injury; *BP,* blood pressure; *CKD,* chronic kidney disease; *CNS,* central nervous system; *CVD,* cardiovascular disease.

at term,[15] supporting the use of 140/90 mm Hg as the upper threshold for normal BP in pregnancy. The different hypertensive disorders of pregnancy are shown in Fig. 44.3.

Gestational hypertension is the development of new hypertension after 20 weeks' gestation in the absence of features of preeclampsia. Approximately 25% will progress to develop preeclampsia. Gestational hypertension usually resolves by 12 weeks postpartum. If hypertension persists beyond that, the possibility of chronic hypertension should be considered.

Preeclampsia is new hypertension after 20 weeks' gestation either with maternal organ dysfunction (proteinuria, AKI, elevated transaminases, cerebral, hepatic or clotting abnormalities) or uteroplacental dysfunction (fetal growth restriction, abnormal umbilical artery Doppler waveform, or stillbirth). Proteinuria (uPCR >30 mg/mmol or uACR >8 mg/mmol) is not essential for the diagnosis of preeclampsia if other criteria exist[42] (Table 44.4).

Eclampsia is seizure activity in a woman with preeclampsia. It is uncommon in high-income countries, with a prevalence of about 0.3% of hypertensive pregnancies. In low- and middle-income countries, eclampsia is more common, with greater risks for maternal mortality and morbidity, as well as perinatal mortality.

Chronic hypertension is BP 140/90 mm Hg or higher that predates pregnancy or is detected before the 20th week of pregnancy on at least two occasions and persists beyond 12 weeks postpartum. This can be confirmed by home BP monitoring or 24-hr ambulatory BP monitoring. Superimposed preeclampsia is the new development of features outlined in Table 44.4 after 20 weeks' gestation in a woman with chronic hypertension.

White-coat hypertension is elevated office or clinic BP (≥140/90 mm Hg) but normal BP measured at home (<135/85 mm Hg). It is not an entirely benign condition with an increased (8%) risk of preeclampsia, which is about double the usual risk.

Epidemiology

Hypertension affects 10% to 12% of all pregnancies, and the global estimate for the incidence of preeclampsia is 4.6% of all pregnancies, though there is variation due to patient factors such as ethnicity and vascular risk, as well as nonstandardization of diagnostic criteria. The incidence of eclampsia is 1.4% of pregnancies globally, though this varies between countries from 0.02% to 2.9%. In low- and middle-income countries, mortality from preeclampsia accounts for nearly 30% of maternal deaths. Such mortality rates are 200 times higher than those of high-income countries that have universal access to health care, consistently applied national guidance for management, and a low threshold for the treatment of severe hypertension in pregnancy.[42] Chronic hypertension is estimated to affect 1% of pregnancies, which may increase due to advancing maternal age and rising rates of pregestational diabetes, obesity, and metabolic syndrome. Black, Asian, Hispanic, American Indian, Alaskan Native, and indigenous Australian populations have a higher risk for preeclampsia than non-Hispanic Whites, with a higher prevalence of hypertension, obesity, and diabetes contributing.

PREECLAMPSIA

Box 44.2 lists known risk factors for preeclampsia. The risk is highest in those with a history of preeclampsia, ranging from 15% to 65% depending on the gestation at which preeclampsia developed in a previous pregnancy. Early development of preeclampsia requiring delivery before 34 weeks is associated with the highest risk of recurrence. Antiphospholipid syndrome, diabetes, obesity, chronic hypertension, CKD, and assisted reproduction all confer an increased risk of preeclampsia.[43] In general, the risk of preeclampsia is higher in a first pregnancy compared with subsequent pregnancies. However, risk returns to that of a first pregnancy in women who have a new partner or an interpregnancy interval beyond 7 years.

TABLE 44.4 Diagnostic Criteria for Preeclampsia

Essential criteria	>20 weeks' gestation *and* New hypertension: systolic BP ≥140 mm Hg or diastolic BP ≥90 mm Hg on two occasions
Additional criteria	Proteinuria: • uPCR >30 mg/mmol (≥0.3 mg/mg) • >300 mg/24 h (not indicated if uPCR available) • Dipstick >2+ (if other methods unavailable) • uACR >8 mg/mmol (>70 mg/g) Serum creatinine: • An increase of serum creatinine to >1.0–1.1 mg/dL (>90–100 µmol/L)[a] • Doubling of serum creatinine in pregnancy (even if new serum creatinine <1.1 mg/dL)[a] Hematologic complications: • Platelets <150 × 10^9/L[a] • Hemolysis,[a] disseminated intravascular coagulation[a] Liver complications: • AST or ALT >40 U/L or double normal reference limit[a] • Epigastric/right upper quadrant pain (not attributable to alternate diagnosis)[a] Neurologic complications: • Eclampsia[a] • Altered mental status[a] • Blindness, persistent visual scotomata[a] • Stroke[a] • Clonus • New onset headache not attributable to alternate diagnosis[a] Respiratory complications: • Pulmonary edema[a] Uteroplacental dysfunction: • Fetal growth restriction, abnormal umbilical artery • Doppler waveform, stillbirth

Diagnosis requires one essential and one additional clinical feature for diagnosis.

[a]Severe features.

ALT, Serum alanine transferase; *AST,* serum aspartate transferase; *BP,* blood pressure; *uACR,* urinary albumin:creatinine ratio; *uPCR,* urinary protein:creatinine ratio.

Modified from Brown MA, Magee LA, Kenny LC, et al; International Society for the Study of Hypertension in Pregnancy (ISSHP). Hypertensive disorders of pregnancy: ISSHP classification, diagnosis, and management recommendations for international practice. *Hypertension.* 2018;72:24–43; ACOG practice bulletin no. 202 summary: gestational hypertension and preeclampsia. *Obstet Gynecol.* 2019;133:211–214; and Wiles K, Chappell LC, Lightstone L, Bramham K. Updates in diagnosis and management of preeclampsia in women with CKD. *Clin J Am Soc Nephrol.* 2020;15:1371–1380.

BOX 44.2 Risk Factors for Preeclampsia

Maternal Factors

Obstetric

- Nulliparity
- Multiple-gestation pregnancy
- History of previous preeclampsia
- Prior intrauterine growth restriction
- Prior placental abruption
- Prior stillbirth
- Artificial reproductive technology
- Molar pregnancy
- Trisomy 13 or fetal hydrops
- Gestational diabetes
- First trimester biophysical/biochemical markers: uterine artery pulsatility index, uterine mean arterial pressure, serum pregnancy-associated plasma protein A and placental growth factor[a]

Comorbidity

- Chronic hypertension
- Chronic kidney disease
- Pregestational diabetes
- Prepregnancy body mass index >30 kg/m^2
- Antiphospholipid antibody syndrome
- Systemic lupus erythematosus
- Polycystic ovarian syndrome

Genetic

- Thrombophilia
- Preeclampsia in first-degree relative

Other

- Age >40 years
- Having been born small for gestational age

Paternal Factors

- New partner
- Conception by intracytoplasmic sperm injection
- Father born from pregnancy affected by preeclampsia

[a] Hofmeyr GJ, Lawrie TA, Atallah AN, et al. Calcium supplementation during pregnancy for preventing blood pressure disorders and related problems. *Cochrane Database Syst Rev.* 2018;10(10):CD001059.

Pathogenesis

The placenta is likely to be key in the development of preeclampsia, secreting factors that cause systemic endothelial dysfunction (Fig. 44.4).[44] Hence, preeclampsia can occur in hydatidiform mole, where the fetus is absent, and it resolves after delivery and removal of the placenta.

Normal placental development is characterized by remodeling of the uterine spiral arteries within the endometrium that supply the placenta. The process is driven by cytotrophoblasts derived from the conceptus, which surround the walls of the spiral arteries, leading to destruction of smooth muscle and elastin within the arterial wall and replacement with inert fibrinoid material, thereby preventing vasoactivity and vasoconstriction. This conversion of the spiral arteries produces capacitance vessels capable of carrying blood at a reduced velocity and pulsatility to placental villi, allowing sufficient oxygen exchange between mother and fetus.

Although deficient spiral artery remodeling has also been described in normal pregnancies and in fetal growth restriction in the absence of preeclampsia, there is consensus that this finding is most marked in preeclamptic pregnancies. A failure of spiral artery remodeling results in uteroplacental ischemia (though this may not be the only mechanism producing an hypoxic placental environment), indicated by the finding of hypoxia-inducible factor 1α, a marker of cellular oxygen deprivation, in the placentas of women with preeclampsia.[45] Oxidative stress, activation of nuclear factor-κB, increased necrotic trophoblast shedding, and proinflammatory interleukin production have all been mechanistically linked to the hypoxic microenvironment. Pathologic inflammation is also evidenced by acute atheroma, which is seen in 10% of preeclamptic placentas.[46]

Placental hypoxia leads to an altered balance between angiogenic (placental growth factor [PlGF], vascular endothelial growth

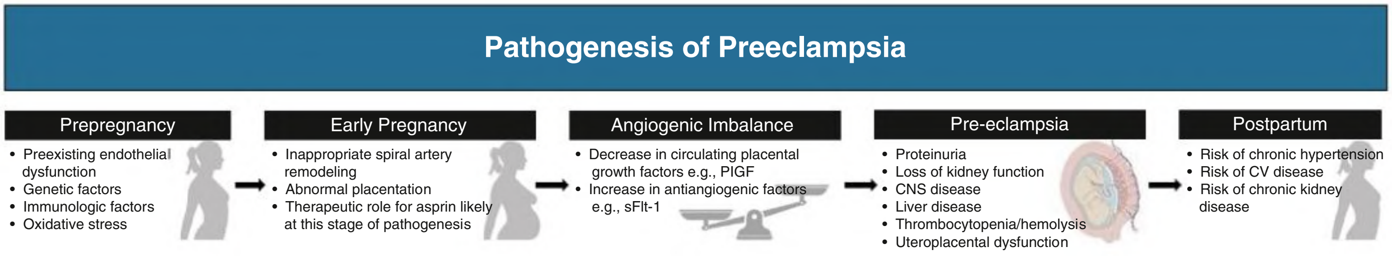

Fig. 44.4 **Pathogenesis of Preeclampsia.** Prepregnancy risk factors and hypothesized pathologic mechanisms contribute to the failure of trophoblast uterine interactions in early pregnancy. Stress to the syncytiotrophoblast leads to the release of a range of factors into the systemic circulation producing an imbalance between proangiogenic factors including placental growth factor (PIGF) and antiangiogenic factors including soluble fms-like tyrosine kinase-1 (sFlt-1), which is detectable before the clinical syndrome of preeclampsia manifests. Maternal endothelium dysfunction produces the clinical syndrome of preeclampsia. Preeclampsia confers a risk of chronic hypertension, and cardiovascular (CV) and kidney disease later in life. (Pre-eclampsia image from Maher GM, McCarthy FP, McCarthy CM et al. A perspective on pre-eclampsia and neurodevelopmental outcomes in the offspring: Does maternal inflammation play a role? Int J Dev Neurosci. 2019;77:69–76.)

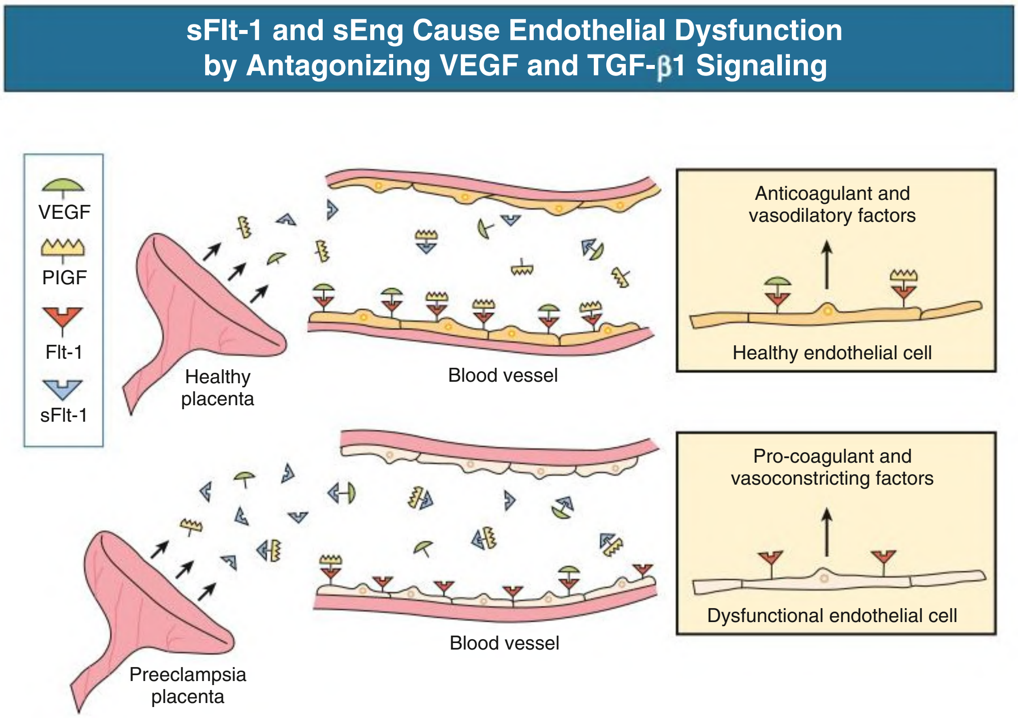

Fig. 44.5 **Proteins Soluble fms-like Tyrosine Kinase 1 (sFlt-1) and Soluble Endoglin (sEng) Cause Endothelial Dysfunction by Antagonizing Vascular Endothelial Growth Factor (VEGF), Placental Growth Factor (PIGF), and Transforming Growth Factor β1 (TGF-β1) Signaling.** There is mounting evidence that VEGF and TGF-β1 are required to maintain endothelial health in several tissues, including the kidney and perhaps the placenta. During normal pregnancy, vascular homeostasis is maintained by physiologic levels of VEGF and TGF-β1 signaling in the vasculature. In preeclampsia, excess placental secretion of sFlt-1 and sEng (two endogenous circulating antiangiogenic proteins) inhibits VEGF/PIGF and TGF-β1 signaling, respectively. This results in endothelial cell dysfunction, including decreased prostacyclin, nitric oxide production, and release of procoagulant proteins. *sFlt-1,* Soluble Fms-like tyrosine kinase 1; *PIGF,* placental growth factor. (Modified from Karumanchi SA, Epstein FH. Placental ischemia and soluble fms-like tyrosine kinase 1: cause or consequence of preeclampsia? *Kidney Int.* 2007;71[10]:959961.)

factor [VEGF], transforming growth factor beta [TGFβ]) and antiangiogenic (soluble fms-like tyrosine kinase 1 [sFlt-1] and soluble endoglin [sEng]) factors, which are measurable in the maternal circulation. sFlt-1 and sEng are syncytiotrophoblast-derived antiangiogenic factors, which bind to PlGF, VEGF, and TGFβ, preventing their interaction with endothelial receptors, thereby inducing endothelial dysfunction, with increased sensitivity to proinflammatory factors and inhibition of vasodilatory pathways (Fig. 44.5). Thus, insufficient placental function, the release of antiangiogenic placental factors into the maternal circulation, and an exaggerated maternal inflammatory response produce the preeclamptic phenotype of generalized endothelial dysfunction including clotting activation, glomerular endotheliosis, transaminitis, and cerebral and pulmonary edema.

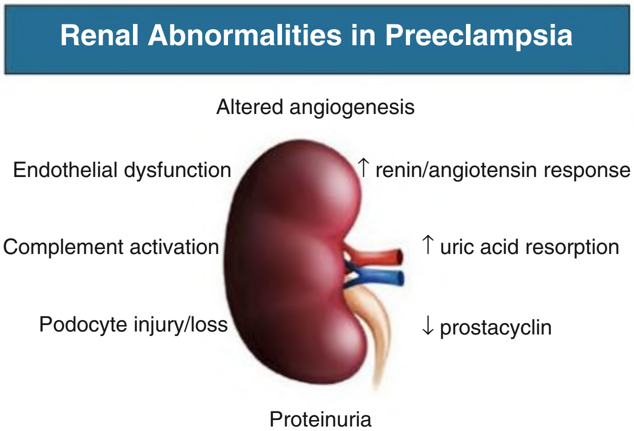

Fig. 44.6 Renal abnormalities in preeclampsia. *GFR,* Glomerular filtration rate. (Modified from Hoenig MP, Hladik GA. Overview of kidney structure and function. In: *National Kideny Foundation Primer on Kidney Diseases,* ed 8. 2022:2–18.)

The hypertensive phenotype can be explained by an exaggerated pressor response, possibly mediated by angiotensin II type 1 receptor (AT_1) autoantibodies. These antibodies are detected in some women with preeclampsia and can activate the AT_1 receptor, produce reactive oxygen species (ROS), increase thrombin generation, impair fibrinolysis, and cause endothelial cell damage.[45] However, the importance of AT_1 autoantibodies in the pathogenesis of preeclampsia remains to be determined.

The clinical presentation of preeclampsia depends on how maternal organ systems and the fetus are affected, but once begun, preeclampsia runs a progressive course until removal of the placenta at delivery, which remains the only definitive cure.

Kidney Abnormalities in Preeclampsia

Histologic changes in the kidney include diffuse *glomerular endotheliosis,* characterized by swelling and vacuolization of endothelial cells, capillary lumen occlusion, and glomerular enlargement (Fig. 44.6). The swollen endothelial cytoplasm encroaches on glomerular capillary lumina, contributing to tuft ischemia (Fig. 44.7). Immunofluorescence

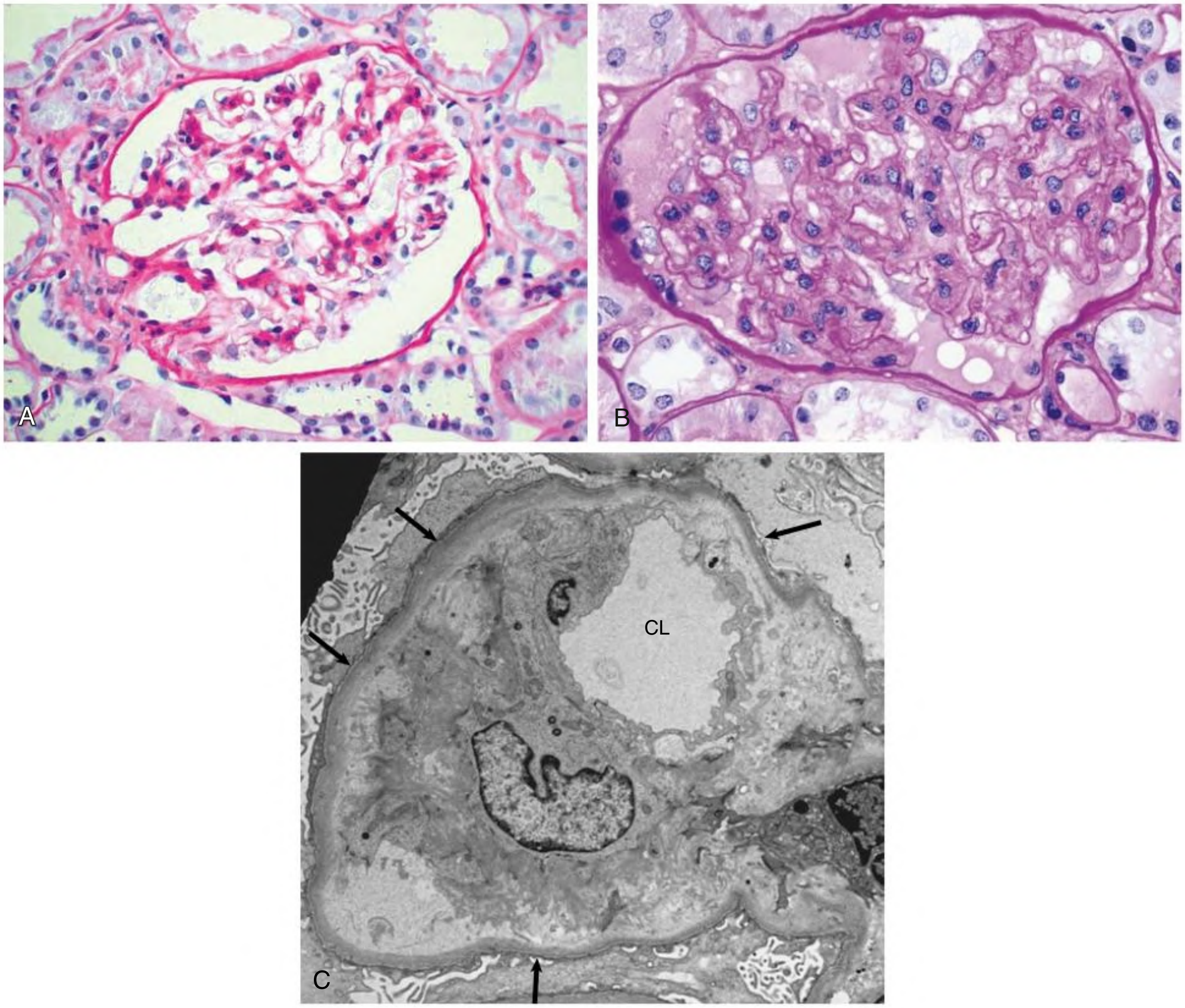

Fig. 44.7 Glomerular Endotheliosis. (A) Normal glomerulus on light microscopy. (B) Glomerulus from a patient with preeclampsia on light microscopy. Note occlusion of capillary lumina by swollen endothelial cells. (C) On electron microscopy, note glomerular basement membrane *(arrows)* and marked reduction of capillary lumen *(CL)* caused by swollen endothelial cell cytoplasm. (A–B, Periodic acid–Schiff reaction; magnification ×40; C, original magnification ×7500.) (Courtesy Prof. P. Furness, University of Leicester, UK.)

may reveal fibrin deposits. However, preeclampsia is a clinical diagnosis and kidney biopsy is not indicated.

Proteinuria

Proteinuria in preeclampsia may be part of the general capillary leak of preeclampsia or a consequence of glomerular endotheliosis. Both glomerular and tubular patterns of proteinuria are reported. Glomerular proteinuria is nonselective and varies from the diagnostic threshold of 300 mg/day up to the nephrotic range. Levels of proteinuria in women with preeclampsia do not consistently correlate with maternal and fetal outcomes.[47,48]

Decreased Glomerular Filtration Rate

GFR is decreased by approximately one-third in women with preeclampsia compared with normotensive late pregnancy. This reduction in GFR is greater than the measured reduction in blood flow, leading to the conclusion that glomerular endotheliosis is an important causative factor in addition to the hemodynamic changes of preeclampsia, including vasoconstriction, plasma volume contraction, and altered cardiac output.

Sodium Retention

Sodium retention occurs in preeclampsia, thought to be a renal tubular response to the perceived reduction in kidney perfusion, increased sympathetic nervous system activity, and/or altered expression of epithelial sodium channels.

Increased Uric Acid Reabsorption

Hyperuricemia in preeclampsia results primarily from increased renal uric acid retention in the kidney, perhaps with increased placental production. Though hyperuricemia has been correlated with fetal risk, serum uric acid concentrations are not a diagnostic criterion for preeclampsia.

Clinical Manifestations

A diagnosis of preeclampsia requires a combination of new onset of hypertension after the 20th week of pregnancy, with evidence of maternal endothelial dysfunction or uteroplacental compromise (see Table 44.4). Symptoms are not always present but may include headache, visual scotomata, seizures, focal neurologic deficit (manifestations of cerebral involvement), epigastric or right upper quadrant pain (reflecting hepatic disease), oliguria, abdominal pain (placental abruption), or reduced fetal movements (Box 44.3). Neurologic complications include seizures (eclampsia), ischemic and hemorrhagic stroke, retinal detachment, cortical blindness, and posterior reversible encephalopathy. Hepatic complications include transaminitis, subcapsular hematoma, and in rare cases liver rupture. AKI occurs, though requirement for dialysis is rare in the absence of significant hemodynamic complications at the time of delivery, or severe disease with thrombotic microangiopathy. Cardiorespiratory complications include myocardial ischemia and pulmonary edema.

Diagnosis and surveillance of preeclampsia require the following:

- Quantitation of proteinuria by uPCR or uACR (diagnostic only)
- Hemoglobin and platelet count
- Serum creatinine and electrolytes
- Serum transaminases
- Ultrasound assessment of fetal growth, amniotic fluid volume, and umbilical artery flow

Routine evaluation in preeclampsia should include BP measurement, assessment for pulmonary edema, assessment for clonus as a warning sign of impending eclampsia, fetal heart rate patterns measured by cardiotocography (CTG) and assessment of fetal growth.

BOX 44.3 Clinical Features of Preeclampsia

Primary Manifestation

- Hypertension

Kidney Involvement

- Significant proteinuria: spot ratio of urine protein to creatinine >30 mg/mmol (≥0.3 mg protein/mg creatinine)
- Acute kidney injury: serum creatinine >1.0–1.1 mg/dL (>90 mmol/L)
- Oliguria

Hematologic Involvement

- Thrombocytopenia
- Hemolysis
- Disseminated intravascular coagulation

Liver Involvement

- Raised serum transaminases
- Severe epigastric or right upper quadrant pain

Neurologic Involvement

- Seizures (eclampsia)
- Hyperreflexia with sustained clonus
- Severe headache
- Persistent visual disturbances (photopsia, scotomata, cortical blindness, retinal vasospasm)
- Cerebrovascular accident (stroke)

Other Major Features

- Pulmonary edema
- Fetal growth restriction
- Placental abruption

Eclampsia

Eclampsia is the occurrence of tonic-clonic seizures in a woman with preeclampsia. The majority, but not all, have premonitory signs and symptoms in the week before a first eclamptic seizure including headache (56%), visual disturbances (23%), epigastric pain (17%), hypertension (48%), proteinuria (46%), or concurrent hypertension and proteinuria (38%).[49] Risk for eclampsia is not directly related to the level of BP, and it can occur in treated hypertension and in the absence of proteinuria. Importantly, about half the cases of eclampsia occur after delivery, usually within the first 5 postpartum days.

HELLP Syndrome

HELLP syndrome (hemolysis, elevated liver enzymes, and low platelet count) is considered a severe form of preeclampsia in which hepatic and platelet abnormalities dominate, with thrombotic microangiopathy. Proteinuria may be absent. The diagnosis of HELLP syndrome is based on the following laboratory criteria:

- Microangiopathic hemolytic anemia: schistocytes (fragments) in a blood film, elevated serum bilirubin, high lactate dehydrogenase, low haptoglobin
- Increased liver transaminases: aspartate transaminase greater than 70 U/L or more than twice the upper limit of normal
- Platelet count: less than 100×10^9/L

Maternal mortality from severe HELLP syndrome is about 1% and perinatal mortality between 7% and 34%, largely depending on gestational age.[50] It is advisable to treat this as a severe variant of preeclampsia, for which iatrogenic delivery is usually indicated to prevent morbidity and mortality. Maternal steroid administration is therapeutic and is not recommended.

Natural History

In general, the prognosis is favorable for women with preeclampsia, but complications include abnormal liver function or thrombocytopenia (10%–20%), pulmonary edema (<0.5%), AKI (1%–5%), placental abruption (1%–4%), fetal growth restriction (10%–25%), neurologic damage (<1%), preterm birth, and perinatal death.[51] After delivery, clinical and laboratory derangements of preeclampsia resolve, but this may be delayed in some women, and it is critical to remain vigilant until there is definite improvement in the clinical picture and hematologic and biochemical parameters. Some patients require antihypertensives for the first time in the postnatal period, which we recommend for BPs in excess of 140/90 mm Hg. Hypertension may persist for days, weeks, or months but will eventually resolve if preeclampsia is responsible for the elevated BP and there is no underlying chronic hypertension.

Prediction

Combination screening, including maternal risk factors, late first-trimester uterine artery Doppler ultrasound, and various biomarkers, has been shown in some studies to predict early-onset preeclampsia with high sensitivity and specificity,[52,53] though clinical validation is awaited. The use of a combined first-trimester screen has shown that the detection of preeclampsia requiring delivery before 34 weeks' gestation, which is associated with the highest neonatal morbidity and mortality, can be predicted in 50% of cases using maternal characteristics, improved to 75% by the addition of biochemical markers (serum pregnancy-associated plasma protein-A and placental growth factor), to 90% by addition of biophysical markers (uterine artery pulsatility index, uterine mean arterial pressure), and to 95% if biochemical and biophysical markers are both used.[54] This strategy was used to detect high-risk women in a randomized controlled trial (RCT) of aspirin for the prevention of preeclampsia, with a detection rate of 77%.[55] Such screening may offer clinical utility for women who would not qualify for aspirin prophylaxis based on maternal characteristics alone, provided it is affordable, where standards in measuring pulsatility indices are met, and where appropriate counseling is available.

Changes in angiogenic (lower PlGF) and antiangiogenic (higher sFlt-1) factors are also predictive in preeclampsia. A PlGF concentration lower than 100 pg/mL before 35 weeks' gestation has been shown to rule out the need for delivery due to preeclampsia within the next 2 weeks with 98% probability.[56] Similarly, a ratio of sFlt-1:PlGF lower than 38 has a negative predictive value of 99% for the development of preeclampsia within 1 week and 95% for the next 4 weeks.[57] One RCT of revealed versus concealed PlGF testing showed that PlGF testing in suspected preeclampsia reduced the time to diagnosis and maternal adverse event rate, with no difference in perinatal outcomes.[58] There has been variable adoption of these markers in the assessment of women with placental insufficiency, including suspected preeclampsia, with the outcome of further studies awaited.

Prevention

The observation that preeclampsia is associated with increased platelet turnover and increased platelet-derived thromboxane levels led to several trials investigating the effect of aspirin in the prevention of preeclampsia. Low-dose aspirin (75–150 mg daily) has been shown to reduce the incidence of preeclampsia.[59-62] A recent RCT involving high-risk women was based on maternal risk characteristics and biophysical and biochemical markers. The trial, in which women were given 150 mg aspirin daily versus placebo for 11 to 14 weeks, demonstrated a 62% reduction in preterm preeclampsia.[63] We advocate starting aspirin in all high-risk women at a dose of 150 mg/day in early pregnancy (by 16 weeks' gestation) until 36 weeks.

Evidence that high-dose oral calcium supplementation reduces the risk of preeclampsia is of low quality.[64] The World Health Organization recommends calcium supplementation (1.5–2 g/day) after 20 weeks' gestation only in women with low dietary calcium intake, and calcium supplementation is not given routinely in most high-income countries. Nutritional supplements, including magnesium, folic acid, fish oils, antioxidants, and garlic, have no efficacy in preventing preeclampsia.[65] In particular, combined use of vitamins C and E has been associated with worse maternal and fetal outcomes. Anticoagulation is not recommended for reducing the risk of preeclampsia. Maternal obesity is associated with an increased risk for preeclampsia, and weight loss is recommended to reduce risk.

Management

All women with preeclampsia should be reviewed at the initial diagnosis to assess the severity of the condition and the well-being of the fetus. In some centers, PlGF-based testing is used to predict the risk of preterm delivery and serious maternal complications to guide decisions about the appropriate place of care. If BP is controlled in the absence of severe features, fetal compromise or other clinical concern, outpatient care may be appropriate provided resources, expertise, and regular repeat surveillance are available. Face-to-face clinical reviews should take place at least twice weekly, including platelet count, serum creatinine, liver function, and CTG as clinically indicated. Repeated quantification of proteinuria is not required once the protein-to-creatinine ratio has reached the diagnostic threshold of 30 mg/mmol, because the amount or rate of increase in proteinuria does not reliably predict adverse maternal or perinatal outcome. Ultrasound estimation of fetal growth, amniotic fluid volume, and umbilical artery blood flow should be done at diagnosis and at least every 2 weeks, though a higher frequency may be indicated according to the degree of uteroplacental dysfunction seen.

Blood Pressure Management

The main indication for antihypertensive therapy in preeclampsia is to prevent maternal severe hypertension with risk of intracerebral hemorrhage and placental abruption. Treatment of hypertension does not affect the disease course of preeclampsia, which continues to be driven by irreversible, abnormal placental function. However, control of maternal BP may allow pregnancy to be prolonged, with further fetal maturation.

The BP threshold at which antihypertensive treatment should be initiated is controversial. An RCT of "tight" (target diastolic 90 mm Hg) with "less tight" (target diastolic 100 mm Hg) BP control in pregnancy found that although the achieved BP differences between groups were less than intended (139/90 mm Hg vs. 133/85 mm Hg), tighter BP control was associated with a reduced incidence of severe maternal hypertension (28% vs. 41%) without fetal compromise.[66] A secondary analysis showed that in women with less tight BP, severe hypertension was associated with an increased risk of adverse maternal and fetal outcomes.[67] A systematic review of 59 trials and 4723 women with BPs in pregnancy between 140/90 and 169/109 mm Hg concluded that antihypertensive treatment halves the risk of severe maternal hypertension, with no adverse effects on maternal and fetal outcomes.[68] Such studies have led to a fall in the treatment targets for hypertension in pregnancy over recent years (Table 44.5). An outlier in this trend is the United States, where antihypertensive treatment is not recommended unless BP is higher than 160/110 mm Hg for women with gestational hypertension or preeclampsia[69] and 150/100 mm Hg in women with chronic hypertension and end organ damage.[70] Falling rates of maternal death from hypertensive disorders in the United Kingdom contrast with

TABLE 44.5 Recommended Blood Pressure Targets in Pregnancy

Professional Society	Blood Pressure for Treatment Initiation (mm Hg)	Treatment Target (mm Hg)
NICE, 2010	>150/100 >140/90 if target organ damage	<150/80–100
ACOG, 2013	≥160/110 ≥160/105 if chronic hypertension	120–160/80–105 for chronic BP
SOGC, 2014	>160/110 140–159/90–109 if comorbid conditions	130–155/80–105 <140/90 if comorbid conditions
SOMANZ, 2014	≥160/110 for mild/moderate BP ≥170/110 for severe BP	NA
ISSHP, 2014	160–170/110 for preeclampsia	NA
ISSHP, 2018	>140/90 (>135/85 home)	110–140/85 Reduce medication if diastolic <80
NICE, 2019	>140/90	110–135/70–85
ACOG, 2020	≥160/110 ≥150/100 if end organ damage	120–160/80–110 if chronic hypertension

Over the past 10 years, there has been a trend to reduce the target blood pressure in pregnancy due to evidence of maternal benefit in the absence of fetal harm.

ACOG, American College of Obstetricians and Gynecologists; *ISSHP,* International Society for the Study of Hypertension in Pregnancy; *NA,* not applicable; *NICE,* National Institute for Health and Care Excellence; *SOGC,* Society of Obstetricians and Gynaecologists of Canada; *SOMANZ,* Society of Obstetric Medicine of Australia and New Zealand.

global rates and those in the United States. This has been attributed to strict management protocols that include adequate treatment of hypertension to increasingly lower BP targets in addition to avoiding fluid overload, seizure prophylaxis, and planned delivery after 36 ± 6 weeks' gestation.[71] Our practice is to target treatment to achieve a BP of 110/70 to 135/85 mm Hg in pregnancy.

The choice of antihypertensive agent in pregnancy is determined by licensing, availability, and clinician experience, with no high-level evidence to guide prescribing. The dosing of antihypertensive medication for pregnancy is shown in Table 44.6. Where urgent blood control is needed, both IV (labetalol, hydralazine) and oral (nifedipine, labetalol, methyldopa)[72] agents can be used. Modified-release oral nifedipine is used in this situation to avoid the precipitous drop in BP that can occur with immediate release preparations before admission/monitoring is established llowing for additional treatment including IV preparations if needed. Fewer safety data are available for amlodipine and doxazosin/prazosin, although no adverse fetal effects are commonly reported. ACE inhibitors and angiotensin receptor antagonists are contraindicated due to fetotoxicity in the second and third trimesters. In the absence of pulmonary edema, diuretics are avoided because of concern regarding further intravascular volume depletion on top of the known volume contraction that exists in preeclampsia.

Eclampsia Prophylaxis and Treatment

Magnesium sulfate loading (4 g over 10–15 minutes) followed by an infusion (1 g/h) is used for eclampsia prophylaxis and treatment. Magnesium is a more effective anticonvulsant in eclampsia than diazepam and phenytoin. When used prophylactically, magnesium halves the risk of eclampsia and therefore likely reduces maternal death.[73] It should be continued for 24 hours after a seizure or until 24 hours after delivery when used as prophylaxis. Magnesium sulfate is excreted by the kidney, so caution is required in women with oliguria, AKI, or CKD. In these settings, we advocate that the full loading dose is given

TABLE 44.6 Antihypertensive Treatment in Pregnancy

Medication	Dose	Notes
Labetalol	Oral: 200–1200 mg/day IV: 50 mg every 10 min to a total 200 mg IV infusion: 20 mg/h, titrated as required	Avoid in asthma, heart block/bradycardia Short duration of action; consider oral dosing three times per day.
Nifedipine	Oral: 10 mg stat dose, 20–90 mg/day	Modified-release preparations are used to prevent a precipitous fall in BP and reflex tachycardia. Sublingual preparations are not recommended. Doses can be repeated after 30–60 min. May cause headache and peripheral edema.
Methyldopa	Oral: 250–750 mg three times per day	Wean after delivery due to exacerbation of postnatal depression. 1000 mg orally has been used in the treatment of severe hypertension, with less effect compared with nifedipine.[a]
Hydralazine	Slow IV: 5 mg over 5–10 min, repeated after 20–30 min Oral: 25–50 mg three times/day	IV is used for control of severe hypertension where labetalol is contraindicated or ineffective. Monitor for hypotension and tachycardia. Preloading with IV fluid may be indicated if volume deplete on clinical assessment (250 mL and review). Rarely used orally for acute BP lowering due to unpredictable hypotensive effect.
Clonidine	Oral: 0.05 mg two to three times per day, maximum 1.2 mg/day	
Prazosin	Oral: 0.5–5 mg once to three times/day	Can cause orthostatic hypotension Limited safety data
Amlodipine	Oral: 5–10 mg/day	Limited safety data

Oral dosing is recommended where adequate control can be achieved and expectant management is ongoing. IV dosing is recommended for urgent control in severe hypertension (>160/110 mm Hg), where oral agents do not achieve the desired clinical effect or for precise control at the time of delivery.

[a]Easterling T, Mundle S, Bracken H, et al. Oral antihypertensive regimens (nifedipine retard, labetalol, and methyldopa) for management of severe hypertension in pregnancy: an open-label, randomised controlled trial. *Lancet.* 2019;394:1011–1021.

BP, Blood pressure; *IV,* intravenous.

but the concentration of the ongoing infusion is reduced. Though a serum concentration 1.8 to 3.0 mmol/L has been suggested as therapeutic, monitoring is controversial as concentrations do not correlate with clinical toxicity. In the absence of kidney impairment, we do not routinely measure serum magnesium concentrations. Assessment of respiratory rate, oxygen saturation, examination for hyporeflexia, and electrocardiogram review are required. In our view, seizure prophylaxis should be given to women with preeclampsia and severe features, particularly if there is clinical evidence of cerebral involvement including clonus, severe headaches, and visual scotomata. These women should simultaneously have a plan made for delivery. Withholding seizure prophylaxis in women with preeclampsia without severe features is associated with an extremely low likelihood of seizure and avoids the potential for drug toxicity, which occurs in up to 25% of women treated with magnesium sulfate.[73,74]

Fluid Balance

Preeclampsia is a volume-contracted, not volume-depleted, state. In addition, increased capillary permeability means that IV volume expansion carries risk, particularly for pulmonary edema. Volume expansion is not warranted in most women with preeclampsia because of the risks of pulmonary edema, and fluid restriction (to ~85 mL/h) in women with preeclampsia with severe features reduces maternal morbidity and mortality. Invasive measures of central pressure are not indicated.

Other

Platelet transfusion is generally given for platelet counts less than 20–30 × 10^9/L to facilitate delivery. Platelet transfusion is avoided at higher counts, as the pathology of thrombocytopenia in preeclampsia is one of platelet consumption, not platelet deficiency. Coagulopathy should also be corrected. Caution is advised in the use of fresh frozen plasma because of the IV volume required, and the use of concentrated preparations should be considered. There is no evidence that plasma exchange or corticosteroids are of therapeutic benefit in HELLP syndrome. Therapeutic adsorption of sFlt-1 by dextran column apheresis is the subject of an ongoing prospective study.[75]

Delivery

The only definitive treatment for preeclampsia is delivery of the placenta. The decision to deliver weighs the risk of progressive maternal disease against the likelihood of complications in the infant due to preterm delivery. Assessment of fetal well-being helps to establish the benefit of further intrauterine time. Timing of delivery should be based on optimizing perinatal outcomes while avoiding maternal risks.

Indications for delivery include the following:

- Progressive evidence of maternal organ dysfunction: worsening kidney or hepatic function, progressive thrombocytopenia, neurologic symptoms or signs
- Inability to control BP
- Failure of fetal growth or concern about fetal status

The Hypertension and Preeclampsia Intervention Trial at Near Term (HYPITAT) showed that after 37 weeks' gestation, the maternal risks of preeclampsia are significantly reduced with delivery, without additional perinatal risks.[76] In contrast, prior to 34 weeks, the advantages of increased fetal maturity mean that, in the absence of the indications listed above, expectant management to prolong pregnancy is the norm. For women with preeclampsia between 34 to 37 weeks' gestation, the situation is less clear. A recent randomized trial of planned delivery versus expectant management for women with preeclampsia between 34 and 37 weeks showed that planned delivery reduced maternal morbidity and severe hypertension compared with expectant management, with three-quarters of expectantly managed women progressing to severe preeclampsia. Although there were more neonatal unit admissions related to prematurity with planned delivery, there was no increase in respiratory or other neonatal morbidity. This suggests that planned delivery for women with preeclampsia can be considered after 34 weeks. The trade-off of a reduced risk of severe hypertension and maternal morbidity, against higher neonatal unit admission without excess morbidity, should be discussed with the mother and the multidisciplinary team involved in her and her baby's care with shared decision-making regarding time of delivery.

Antenatal corticosteroids for fetal lung maturation are routinely given to women who are less than 34 weeks' gestation in whom delivery within the next 7 days is anticipated. Use of corticosteroids for fetal lung maturity after 34 weeks remains controversial with limited prospective, randomized trial data to guide practice. Many consensus protocols do recommend steroids at later gestations, particularly for cesarean delivery, which is associated with higher rates of neonatal respiratory distress compared with vaginal delivery.

The antepartum management of preeclampsia is summarized in Fig. 44.8.

Postpartum Management

Recovery should be anticipated over 5 to 7 days after delivery in most women. In many women, the condition may worsen in the first 3 to 5 days after delivery, and they should continue to be monitored and treated in the same way as antepartum. Persistent hypertension at 3 months postpartum warrants investigation for chronic hypertension. Proteinuria can persist for up to 6 months, though regression would be expected. Nonregression of proteinuria and persistent proteinuria at 6 months postpartum warrant renal referral to exclude the possibility of underlying CKD. Investigation to exclude underlying connective tissue, cardiovascular, kidney, and antiphospholipid disorders should be considered for women with early-onset preeclampsia requiring delivery in the second or early third trimester, before another pregnancy.

Preeclampsia has well-recognized long-term implications, including an increased risk for chronic hypertension, coronary heart disease, stroke, venous thromboembolism, and CKD (Box 44.4). Whether preeclampsia causes, accelerates, or reveals underlying endothelial dysfunction is not clear. It is appropriate to counsel women who develop hypertension in pregnancy on cardiovascular risk factor lifestyle modifications, along with interval BP, serum lipid, blood glucose, and kidney function surveillance.

ACUTE FATTY LIVER OF PREGNANCY

Acute fatty liver of pregnancy (AFLP) usually presents in the third trimester and has an incidence of 5 per 100,000 pregnancies,[77] but many milder cases may go unrecognized. Primiparity, twin pregnancies, and low body mass index (<20) are overrepresented in women who develop AFLP compared with the general obstetric population. AKI is a common complication of AFLP.

Pathogenesis and Pathology

AFLP is characterized by lipid microvesicle infiltration of hepatocytes without inflammation or necrosis, though biopsy is not required for clinical diagnosis. AFLP has been linked with rare disorders of mitochondrial fatty acid oxidation, which lead to variable clinical manifestations in those affected including hypoglycemia, hepatic dysfunction, encephalopathy, cardiomyopathy, peripheral neuropathy, and myopathy. Both AFLP and HELLP are described in pregnancies in which the fetus is affected by such a disorder of fatty acid oxidation, of which long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD) deficiency is the best described. The mothers of babies affected by these recessive conditions are default heterozygotes for the same mutation. Although

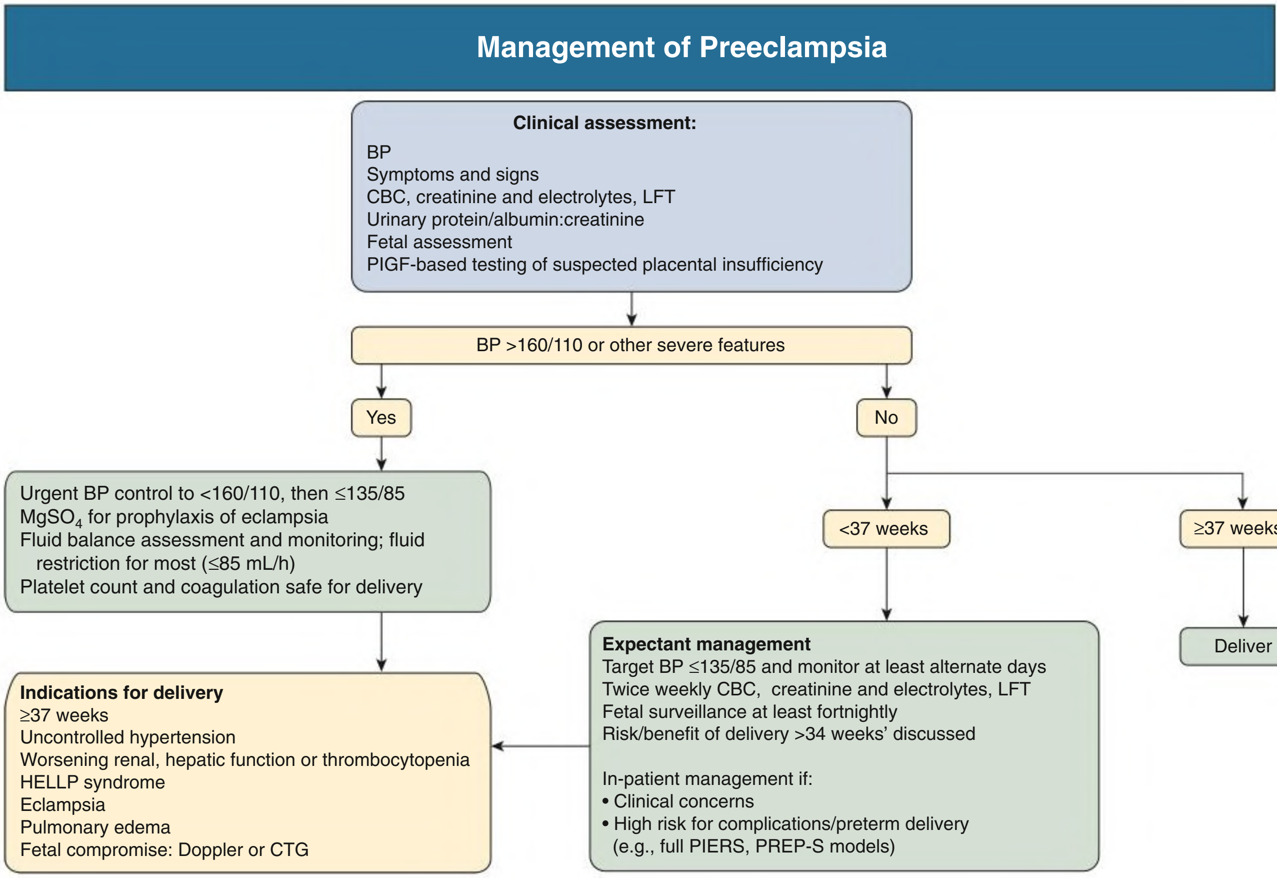

Fig. 44.8 Management of Preeclampsia. Management depends on disease severity and gestational age of diagnosis. *BP,* Blood pressure; *CBC,* complete blood count; *CTG,* cardiotocography; *HELLP,* hemolysis, elevated liver enzymes, and low platelet count; *$MgSO_4$,* magnesium sulfate; *LFT,* liver function test; *PIGF,* placental growth factor. (Modified from Chaiworapongsa T, Chaemsaithong P, Korzeniewski SJ, et al. Pre-eclampsia part 2: prediction, prevention and management. *Nat Rev Nephrol.* 2014;10[9]:531–540.)

BOX 44.4 Long-Term Consequences of Preeclampsia

- Fatal and nonfatal coronary heart disease
- Cerebrovascular accident (stroke)
- Hypertension
- Thromboembolism
- End-stage kidney disease
- Diabetes
- Cognitive dysfunction and white matter lesions on cerebral computed tomography scan
- Death from any cause

heterozygosity is insufficient for disease manifestation outside of pregnancy, the excess fetal fatty acid burden in pregnancy is thought to lead to hepatic deposition and overt maternal liver disease in pregnancy. The screening of infants born after pregnancies affected by AFLP for fatty acid oxidation disorders including LCHAD deficiency has been advocated but is not routine, and prospective data are warranted.[78]

Kidney disease in AFLP is likely to be multifactorial. Kidney biopsy and autopsy studies have shown mild glomerular hypercellularity with thick, narrow capillary loops and tubular free fatty acid accumulation, suggesting that abnormal fatty acid oxidation also may contribute to kidney dysfunction. Hemodynamic changes like those seen in the hepatorenal syndrome, thrombotic microangiopathy, and coexisting preeclampsia may also contribute to kidney injury.

Clinical Manifestations

A diagnosis of AFLP requires six or more clinical criteria (Box 44.5) in the absence of another explanation. The severity of liver involvement in women with AFLP is highly variable, ranging from a moderate isolated increase in transaminases to fulminant hepatic failure with encephalopathy and coagulopathy. Characteristic laboratory abnormalities include hyperbilirubinemia, increased transaminases, hypoglycemia, leukocytosis, and evidence of coagulopathy (hypofibrinogenemia, prolonged prothrombin time, depressed antithrombin III levels).

Differential Diagnosis

Viral hepatitis and biliary obstruction should be excluded. Distinguishing AFLP from HELLP can be challenging because the two disorders share many pathophysiologic and clinical features (Table 44.7). However, new vomiting in late pregnancy, polyuria, coagulopathy, and hypoglycemia increase clinical suspicion of AFLP.

Treatment and Outcome

Early diagnosis, supportive care (reversal of coagulopathy, correction of hypoglycemia and fluid status management), and prompt delivery are critical in the management of AFLP. One-third of cases require admission to intensive care. Kidney disease is described in 14% of women with AFLP, with temporary kidney replacement required in one-quarter of women with AKI.[77] In most women, there is complete liver and kidney recovery after delivery, although progressive liver disease requiring transplantation is described. AFLP has maternal and perinatal mortality rates of 2% and 10%, respectively, in a high-income country with universal access to health care.[77]

THROMBOTIC MICROANGIOPATHY

Pregnancy-associated thrombotic microangiopathy (TMA) is defined by the presence of fibrin and platelet thrombi in the microcirculation, causing organ dysfunction, microangiopathic hemolysis, and thrombocytopenia. HELLP syndrome, atypical hemolytic uremic syndrome (aHUS), and thrombotic thrombocytopenic purpura (TTP) can all cause a pregnancy-associated TMA.

BOX 44.5 Criteria for the Diagnosis of Acute Fatty Liver of Pregnancy

Clinical
Vomiting
Abdominal pain
Polydipsia/polyuria
Encephalopathy

Laboratory Values
Bilirubin >0.8 mg/dL (>14 µmol/L)
Hypoglycemia <70 mg/dL (<4 mmol/L)
Uric acid >5.7 mg/dL (>340 µmol/L)
Leukocytosis >11 × 109/L
ALT or AST >42 U/L
Ammonia >47 µmol/L
Coagulopathy: PT >14 s or APTT >34 s

Other
Ascites or bright liver on ultrasound
Microvesicular steatosis on liver biopsy

Diagnosis requires six or more criteria in the absence of another explanation.
ALT, Alanine aminotransferase; *APTT,* activated partial thromboplastin time; *AST,* aspartate aminotransferase; *PT,* prothrombin time.
Modified from Knight M, Nelson-Piercy C, Kurinczuk JJ, et al. A prospective national study of acute fatty liver of pregnancy in the UK. *Gut.* 2008;57(7):951–956.

Clinical Manifestations and Differential Diagnosis

Diagnosis requires platelet count less than 100×10^9/L, hemoglobin less than 100 g/L, serum lactate dehydrogenase (LDH) more than 1.5 times the upper limit of normal, undetectable serum haptoglobin, a negative direct erythrocyte antiglobulin test, and either the presence of schistocytes (fragments) on blood smear or features of TMA on biopsy of an affected organ.

Women with atypical HUS typically present with AKI, most commonly at the end of pregnancy or in the first 3 months postpartum, although disease is described throughout pregnancy. Although kidney injury predominates in aHUS, cardiac and neurologic disease are also described, so distinction from TTP may be difficult on clinical grounds alone. TTP tends to present in the second and third trimesters at the same time as a physiologic decline in ADAMTS13, the deficiency of which leads to pregnancy-associated TTP.

In practice, it can be difficult to distinguish aHUS and TTP from preeclampsia with severe features (see Table 44.7). Preeclampsia with severe features is a far more common cause of pregnancy-associated TMA than both aHUS and TTP, which are rare. However, recovery of clinical parameters after delivery is expected in preeclampsia/HELLP, and persistent or progressive thrombocytopenia at 48 to 72 hours postpartum should raise clinical suspicion for alternative pathology. Postpartum hemorrhage is another complicating factor. The risk of postpartum hemorrhage is higher in women with thrombocytopenia at the time of delivery, and postpartum hemorrhage is a cause of both TMA and AKI, particularly if there is ischemic necrosis of the kidney cortex. Coagulation studies and liver function tests are usually normal in aHUS and TTP (with the exception of prehepatic hyperbilirubinemia), unlike in HELLP and AFLP. Whether PlGF-based testing offers utility in the distinction of antepartum disease is unknown.

Treatment

Delivery should be considered for all cases of pregnancy-associated TMA, depending on fetal maturity. Delivery is indicated if there is fetal compromise or progressive maternal disease. Delivery is sufficient to control TMA due to preeclampsia/HELLP and may help achieve a better response to treatment in TTP.

Plasma exchange should be commenced in atypical presentations of preeclampsia/HELLP with thrombocytopenia ($<30 \times 10^9$/L) and either life-threatening neurologic features (seizures, altered

TABLE 44.7 Differential Clinical and Laboratory Features of Syndromes That Cause Acute Kidney Injury in Pregnancy

Clinical Features	PE	HUS/TTP	HELLP	AFLP
Hemolytic anemia	+/–	+++	++	+/–
Thrombocytopenia	+/–	+++	++	+/–
Coagulopathy	+/–	–	+/–	+
CNS symptoms	+/–	+/– (HUS) ++ (TTP)	+/–	+/–
Kidney failure	+/–	+++ (HUS)	+	++
Hypertension	++	+/–	+++	+/–
Proteinuria	+/–	+/–	++	+/–
Elevated AST	+/–	+/–	++	+++
Elevated ALT	+/–	–	++	+++
Elevated bilirubin	+/–	++	+	+++
Anemia	+/–	++	+	+/–
Blood ammonia	Normal	Normal	Normal	High
Low ADAMTS13 activity	–	++TTP (but not HUS)	–	–
Effect of delivery on disease	Recovery	None	Recovery	Recovery
Management	Supportive care/delivery	Plasma exchange/eculizumab	Supportive care/delivery	Supportive care/delivery

AFLP, Acute fatty liver of pregnancy; *ALT,* alanine aminotransferase; *AST,* aspartate aminotransferase; *CNS,* central nervous system; *HELLP,* hemolysis, elevated liver enzymes, and low platelet count; *HUS,* hemolytic uremic syndrome; *PE,* preeclampsia; *TTP,* thrombotic thrombocytopenic purpura.

consciousness, coma) or cardiac disease (persistently elevated troponin levels, electrocardiographic abnormalities, impaired cardiac function). Plasma exchange should be continued until ADAMTS13 tests results are available, which will allow confirmation or exclusion of TTP. In the absence of significant neurologic and cardiac disease, parameters can be monitored for postpartum recovery and plasma exchange commenced if anticipated improvements in hemolytic parameters, platelet count, and kidney function are not seen. Plasma exchange is continued only if TTP (or catastrophic antiphospholipid syndrome) is confirmed. If a diagnosis of aHUS is made by exclusion of preeclampsia, short-term treatment with eculizumab can be started and reviewed depending on response to treatment and the results of complement gene testing.[78] Though there are limited data on eculizumab exposure and pregnancy, the benefit in treating organ-threatening disease outweighs risk. Treatment can be ceased if no complement abnormality is demonstrated and clinical recovery occurs quickly after delivery.[33]

ACUTE KIDNEY INJURY

Definition

International criteria for the diagnosis and staging of AKI (e.g., Acute Kidney Injury Network [AKIN]; Risk, Injury, Failure, Loss of Kidney Function, and End-Stage Kidney Disease [RIFLE]; Kidney Disease: Improving Global Outcomes [KDIGO]; see Chapter 72) have not been validated for use in pregnancy. Diagnosis of AKI in pregnancy is therefore based on serum creatinine concentrations. Meta-analysis data show that the upper reference limits (97.5th centile) for serum creatinine in pregnancy are 85%, 80%, and 86% of the female nonpregnant reference range in the first, second, and third trimesters, respectively.[8] This means that for a female reference range for serum creatinine of 45 to 90 μmol/L outside of pregnancy, serum creatinine concentrations greater than 0.86 mg/dL (76 μmol/L), 0.81 mg/dL (72 μmol/L), and 0.87 mg/dL (77 μmol/L) in respective trimesters should trigger investigation for AKI (or undiagnosed CKD).

Epidemiology

AKI is associated with an increased morbidity and mortality regardless of the underlying cause, and it should be considered a serious complication of pregnancy. Audit data demonstrate that AKI complicates 1.4% of obstetric admissions in the United Kingdom. Most cases occur in women without preexisting kidney disease, and it is estimated that more than 40% may be unrecognized by the treating clinical team,[15] masked by gestational changes, the limitations of serum creatinine, and by physiologic oliguria in the immediate postpartum period.

The development of AKI in pregnancy follows a bimodal distribution with two incidence peaks: the first and third trimesters. Prerenal causes are more common in the first trimester because of hyperemesis gravidarum and septic abortion, which remains a common global cause of pregnancy-associated AKI. The most common etiologies in late pregnancy are preeclampsia and hemorrhage.

Etiology and Pathogenesis

The etiology of pregnancy-associated AKI can be considered according to the trimester in which it presents, which aids in establishing an appropriate differential (Table 44.8). Diagnosis may be complicated by the overlapping clinical phenotype of pregnancy-associated conditions. Table 44.7 outlines the differentiating clinical features of preeclampsia, HELLP syndrome, aHUS/TTP, and AFLP.

TABLE 44.8 Causes of Acute Kidney Injury in Pregnancy

Gestation	Diagnosis	Clinical Features
Early	Hyperemesis gravidarum	Nausea, vomiting, ptyalism
	Septic abortion/miscarriage	Abdominal pain, vaginal bleeding, hypotension
	Acute retention	Retroflexed uterus
Mid-late	Preeclampsia/HELLP	New-onset hypertension after 20 weeks' gestation plus evidence of maternal endothelial dysfunction and/or uteroplacental dysfunction
	Ureteral obstruction	Increased risk if single kidney (including transplant), autonomic neuropathy, polyhydramnios, multiple pregnancy, obstructed labor
	Placental abruption	Vaginal bleeding, abdominal pain, uterine tenderness
	AFLP	Vomiting, polydipsia, jaundice, elevated transaminases, hypoglycemia
	Microangiopathic hemolytic anemia (TTP/HUS)	Hemolysis and end organ damage including kidney failure, neurologic symptoms, heart disease
Peripartum	Chorioamnionitis	Tachycardia, fever, uterine tenderness, abnormal lochia, prolonged rupture of membranes
	Postpartum hemorrhage	Hemodynamic instability
	Ureteric injury	Operative delivery, fever, leucocytosis, pain, persistent ileus
	NSAIDs	Hyperkalemia, exacerbation of hypertension, fluid retention
Any	Urosepsis	Dysuria, back or flank pain, renal angle tenderness, hypotension
	Lupus nephritis	Proteinuria ± hematuria, malaise, joint pain, hair loss, rash, pancytopenia
	Glomerulonephritis	uPCR >30 mg/mmol or uACR >8 mg/mmol before 20 weeks warrants nephrology referral
	Interstitial nephritis	New drug exposure including antibiotics, NSAID, proton pump inhibitors, H_2 antagonists; systemic signs can include rash and fever
	Kidney stone disease	Renal colic; increased risk of AKI if single kidney
	Intravascular volume depletion	Sepsis, vomiting, cardiac failure, DKA

AFLP, Acute fatty liver of pregnancy; *AKI,* acute kidney injury; *DKA,* diabetic ketoacidosis; *HELLP,* hemolysis, elevated liver enzymes, and low platelet count; *HUS,* hemolytic uremic syndrome; *NSAID,* nonsteroidal antiinflammatory drug; *TTP,* thrombotic thrombocytopenic purpura *uACR,* urinary albumin/creatinine ratio; *uPCR,* urinary protein/creatinine ratio.

From Wiles KS, Banerjee A. Acute kidney injury in pregnancy and the use of non-steroidal anti-inflammatory drugs. *Obstet Gynaecol.* 2016;18:127–135.

Treatment

Fundamental to the management of AKI is supportive treatment to maintain kidney perfusion. However, assessment of the fluid state of the pregnant patient is complicated by physiologic reductions in blood and oncotic pressure. In addition, young pregnant women can compensate for hypovolemia and hypotension and may present with signs much later than expected. The significance of tachycardia and tachypnea in track and trigger systems, which should be based on parameters for pregnancy,[15] is important. Treatment specific to the etiology of AKI, such as antibiotics in sepsis, should be administered promptly.

If AKI leads to hyperkalemia, calcium salts for cardiac stabilization can be safely given in pregnancy. There are limited data on the use of oral potassium binders (patiromer and sodium zirconium cyclosilicate) in pregnancy, but systemic absorption is limited, so benefit in managing hyperkalemia is likely to outweigh any theoretical risk. Insulin with glucose can be used as a temporizing measure to increase intracellular potassium shift while definitive treatment of AKI is undertaken.

Drug doses may need to be adjusted in AKI, including antibiotics, anticoagulants, insulin, and opiate analgesia. The use of nonsteroidal antiinflammatory drugs, which often form part of a standard postpartum analgesic protocol, should be avoided in women with AKI.

Dialysis

Indications for kidney replacement in pregnancy mirror those in the nonpregnant population and include metabolic acidosis, hyperkalemia, and fluid overload refractory to medical treatment. Uremic symptoms (encephalopathy or pericarditis) are very rarely encountered in pregnancy, as dialysis is usually commenced in women with advanced CKD at lower serum urea concentrations than outside of pregnancy due to the risk of fetotoxicity. It is generally recommended to consider initiation of dialysis at blood urea nitrogen (BUN) greater than 42 mg/dL (serum urea >15 mmol/L) depending on etiology, trajectory, and plans for delivery. Both peritoneal dialysis and hemodialysis have been used in pregnancy. However, there are more data on the use of hemodialysis, which is preferentially used as it offers the capacity to increment dialysis provision as required, aiming for a predialysis BUN less than 35 mg/dL (urea <12.5 mmol/L) in contemporary cohorts.[33] Dialysis in pregnancy is discussed in more detail in Chapter 45.

OVARIAN HYPERSTIMULATION SYNDROME

Definition and Epidemiology

Ovarian hyperstimulation syndrome (OHSS) is an iatrogenic complication of ovarian stimulation therapy, which is used to increase the number of oocytes available for assisted reproductive technology. The clinical picture is of significant fluid shifts detectable as ascites and hemoconcentration. Thromboembolism, kidney injury, and pleural or pericardial effusions represent severe disease. OHSS is estimated to complicate between 0.5% and 5% of stimulatory cycles.

Pathogenesis

OHSS is thought to be due to the release of proinflammatory mediators including VEGF, cytokines, prostaglandins, and histamine, which are produced in response to human chorionic gonadotrophin (hCG) on a background of supraphysiologic ovarian stimulation. These factors increase vascular permeability, resulting in ascites, pleural and pericardial effusions, and edema. Women with severe OHSS are estimated to have lost 20% of their circulating volume.

Clinical Manifestations

Symptoms include abdominal bloating and pain, nausea and vomiting, peripheral edema, oliguria, breathlessness, and orthopnea. Venous thromboembolism can occur. The combination of elevated hematocrit, reduced serum osmolality, and hyponatremia is suggestive of OHSS. Patients should be investigated with full blood count and hematocrit; serum creatinine, electrolytes and osmolality; liver function tests; C-reactive protein; β-hCG; and coagulation profile. Abdominal ultrasound imaging may help clarify the diagnosis. OHSS is not a common cause of fever or peritonitis, in which case ovarian torsion or rupture, intraabdominal infection, and ectopic pregnancy should be excluded.

Management

OHSS is self-limiting, with an average duration of 7 days in nonpregnant women and 10 to 20 days in pregnant women. In the woman with moderate to severe OHSS, hospital admission is required for correcting volume depletion; monitoring and correcting electrolyte abnormalities (particularly hyperkalemia); and providing analgesia, nutrition and psychological support, prophylaxis of venous thromboembolism, and respiratory function support. If IV fluids are necessary, crystalloids are used first line, with close monitoring because of the possibility of third-space accumulation. Paracentesis is sometimes needed to improve symptoms and respiratory function, in which case human albumin solution as a plasma volume expander should be considered. Prophylactic anticoagulation is required because of the risk for thromboembolism.

Women with underlying kidney disease have a higher risk of adverse outcomes in the event of OHSS. In vitro fertilization cycles using gonadotrophin-releasing hormone (GnRH) antagonists have been found to have a lower incidence of OHSS compared with cycles where GnRH agonists are used. Incidence is also higher in cycles when conception occurs compared with cycles without conception, and higher still in cycles resulting in multiple pregnancy, highlighting the important role of endogenous hCG. Stimulation followed by later frozen-embryo transfer may therefore be opted for, rather than fresh-embryo transfer with stimulation and pregnancy in the same cycle.

SELF-ASSESSMENT QUESTIONS

1. Which of the following does not occur in normal pregnancy?
   A. Lower serum osmolality
   B. Hypernatremia
   C. Elevated glomerular filtration rate
   D. Glycosuria
   E. Reduced serum albumin

2. Which of the following is not an indication for delivery in preeclampsia?
   A. Proteinuria 2 g/day
   B. Gestation 37 weeks or more
   C. Pulmonary edema
   D. Rising serum creatinine

E. Hemolysis, elevated liver enzymes, and low platelet count (HELLP)

3. Which one of the following syndromes causing acute kidney injury is least likely to recover after delivery?
   A. Preeclampsia
   B. HELLP syndrome
   C. Hemolytic uremic syndrome
   D. Acute fatty liver of pregnancy
   E. Acute urinary retention
4. Which of the following can be used to predict the need for delivery in suspected preeclampsia?
   A. Proteinuria
   B. Placental growth factor concentrations
   C. Relaxin concentration
   D. Renin concentration
   E. Angiotensin receptor antibody concentration
5. Which of the following can be given in early pregnancy to reduce the risk of preeclampsia?
   A. Folic acid
   B. Vitamin D
   C. Iron
   D. Aspirin
   E. Hydralazine
6. Which of the following is *not* a risk factor for preeclampsia?
   A. First pregnancy
   B. Previous preeclampsia
   C. Interbirth interval less than 1 year
   D. Chronic kidney disease
   E. Antihypertensive requirement before 20 weeks' gestation

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